Ha! Linguistic Studies in Honor of Mark R. Hale



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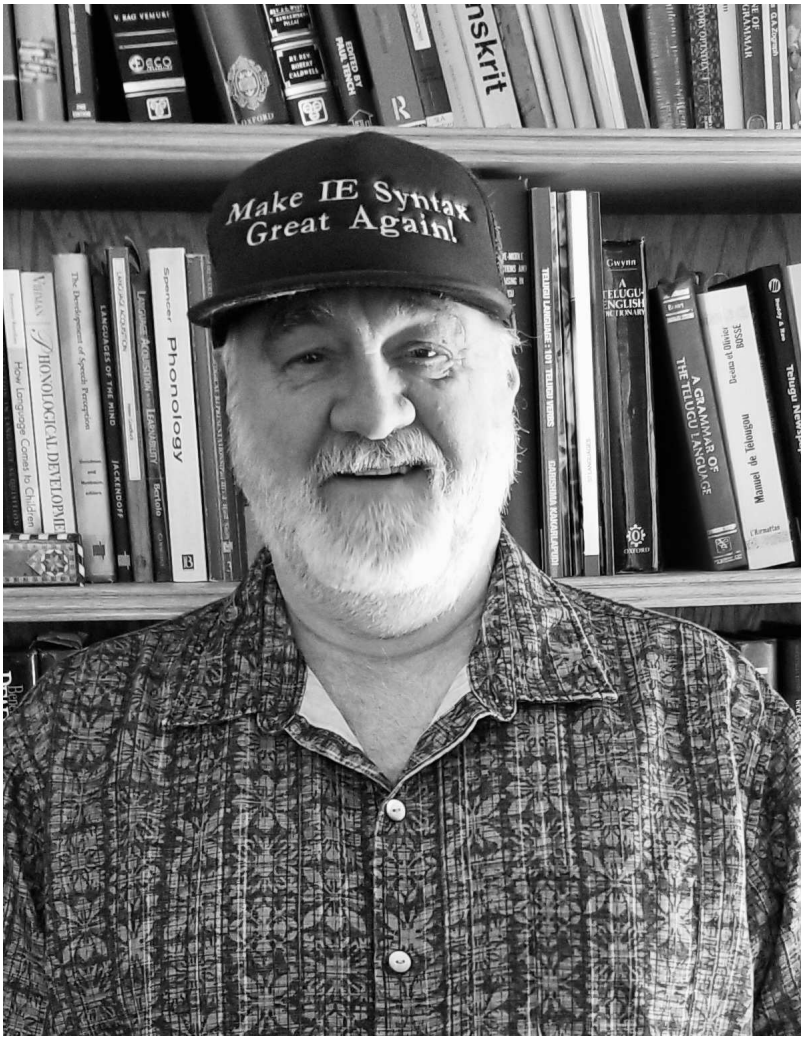
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Latin *glaciēs* 'ice'

Brent Vine

1 Introduction

There is general agreement that the *gl-* of Lat. *glaciēs* f. 'ice' (Varro, Lucr.+) has something to do with the root of Lat. (*u*-stem) *gelus* m. (also neut. *gelū* and *o*-stem *gelum*) 'cold, frost, frosty weather' (Liv. Andr.+) and *gelidus* 'ice cold, frosty' (Cato+). But that is where agreement ends. This apparently intractable situation is well characterized by de Vaan (EDLIL, s.v. *glaciēs*), who observes that "*glaciēs* cannot be derived from a root **ĝl-* 'to be cold' in any meaningful way".[1] Similarly Ernout-Meillet (DELL, s.v. *glaciēs*): "La formation de *glaciēs* n'est pas claire".

There are, however, earlier approaches that deserve comment; these will be examined in section 2. The goal of this modest contribution will be to describe (in section 3) how an updated version of one of these conceptions may provide a workable solution.

2 Earlier approaches

*2.1 PIE *gel- and broken reduplication*

In contrast to the skeptical position taken by Ernout-Meillet and de Vaan, earlier handbooks operate with what may be considered the standard account, going back at least to Fick 1909: 41 (cf. Falk and Torp 1910–1911: 526), and followed by Walde (1927: 1.622), Walde-Hofmann (LEW, s.v. *glaciēs*), and Pokorny (IEW 366). In this approach, the primary comparandum is ON *klaki* m. 'frozen ground, ice', from a Germanic *n*-stem **klak-an-*. This is taken to reflect a PIE form **glag-on-*, showing in **glag-* a so-called "broken reduplication" (i.e., **gla-g-*) based on PIE **gel-* 'cold; be cold', as in the Germanic strong verb seen in ON *kala*, OE *calan* 'become cold, freeze' and words for 'cold' (OHG *kalt* etc.); cf. also OE *ciele* m. 'chill'. Lat. *glaciēs*, then, continues an earlier **glagiēs* (with the same broken reduplication **gla-g-* said to underlie ON *klaki*), adjusted to *glaciēs* after *aciēs* f. 'sharp edge, point' and other Latin words ending in *-aciēs*.[2]

Yet this account raises more questions than it answers. To begin with, the appeal to "broken reduplication"—especially across more than one branch of Indo-European—does not inspire confidence. The episodic nature of this type of structure may be judged from the fact that "broken reduplication", and patterns like it, have virtually no profile in modern typological and theoretical studies of reduplication.[3]

1 Despite de Vaan's notation, the reconstruction with palatal is not certain; properly LIV^2 185, indicating **gel-* or **ĝel-*.

2 The latter set is in fact limited to *faciēs* 'appearance' (Pl.+) and *maciēs* 'thinness' (Lucr.+).

3 See, e.g., Frampton 2009, Inkelas and Downing 2015, Urbanczyk 2017.

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There are, to be sure, analyses of this kind for scattered forms in individual languages.[4] At the level of PIE, however, the best documented instances, such as they are, involve animal terms with a distinctly non-IE appearance, as possibly in 'crab' (**kar-k-*, cf. Gk. καρκίνος and, with dissimilation, Lat. *cancer* < **kar-k-ro-*) and 'ant' (**mor-m-*, cf. Gk. μύρμηξ and, again with dissimilation, Lat. *formīca*).[5] From the perspective of Latin, moreover, isolated *-iēs* forms like *glaciēs* (i.e., not including productive *-itiēs* forms or the type with *-ia* beside *-iēs*) are transparently either root based or associated with related Latin material (e.g., *aciēs* 'sharp edge' [**h₂eḱ-* 'be sharp'], cf. *aceō* 'be sour'; *maciēs* 'thinness', cf. *macer* 'thin'; *speciēs* 'appearance', cf. *speciō* 'look'; etc.);[6] there is nothing comparable to the obscure basis **gla-g-* presumed by this theory. This basis is itself formally problematic: what interpretation is to be placed on the medial *-a-*? Ernout-Meillet (DELL, s.v. *gelū*) address this point by suggesting that *glaciēs* "laisse entrevoir une forme de racine dissyllabique"; and indeed, Pokorny's lemma 3. *gel(ə)-* (IEW 365) points in the same direction. Yet even if the root is reliably reconstructed as laryngeal-final (which is far from certain), there is no phonologically regular method of producing pre-Latin **glă-g-* from full-grade **gelH-g-* (> pre-Latin **gela-g-*) or zero-grade **glH-g-* (> pre-Latin **glā-g-*).[7] On the Germanic side, it is a reasonable supposition, as expressed by Kroonen (EDPG, s.v. *kula-* 1), that for PGmc. **kula-* 'cool wind' (ON *kul*, *kol* n.) "[t]he vocalization of the resonant may point to a root-final laryngeal"; hence Kroonen's reconstruction (with his notation) **ĝlH-o-* for this form, carried over to related material (e.g., **ĝólH-e-* for the *o*-grade present **kalan-* [EDPG, s.v.] of ON *kala*, OE *calan*, cited above). Yet this cannot apply to ON *klaki* itself, the etymology of which is in fact considered by de Vries to be "umstritten" (2000: 313, s.v.), with the old *glaciēs* connection listed third after two other more attractive accounts. In sum, the traditional (pre-Lat.) **gla-g-iēs* approach to Lat. *glaciēs* (based on ON *klaki*) should be abandoned.

*2.2 Compound, PIE *gel- + *h₂eḱ-*

All but forgotten is another approach to Lat. *glaciēs* based on a compound formation involving the root **gel-* (of Lat. *gelus*, *gelidus* etc.) and the root **h₂eḱ-* underlying Lat. *aciēs* 'sharp edge', or Lat. *aciēs* itself: thus for Pictet (1877: 1.113), Lat. *glaciēs* may reflect "une contraction de *gelacies*" (cited with question mark by Vaníček 1881: 83 s.v. GAL); similarly Lewy 1907: 419 fn. 2 ("lat. *glacies* aus **gel-* [*gelu*, *kalt*] + **ak-* 'scharf' ").[8]

There is little need to dilate on this theory, given its evident inadequacies and indeterminacies. Note first that an archaic root-compound would be unique as a basis

4 For Latin itself, see André 1978: 47, 91 and passim.

5 For these reconstructions, see Gamkrelidze and Ivanov 1995: 191.

6 See Weiss 2020: 343 for a convenient survey of the material.

7 The alleged Oscan form identified as γέλα by Stephanus of Byzantium (for the full passage see Billerbeck 2006: 412, and Vetter 1953: 367 s.v. for the Italic context) and glossed as πάχνη 'hoar-frost' says nothing about a root-shape **gelH-* for Italic; and the gloss γελανδρόν· ψυχρόν (Hsch.), which may or may not belong with Gallo-Romance material, is too obscure to be usable for this purpose (see IEW 365 and Beekes, EDG 1.264 s.v. for discussion and references).

8 Rejected by Walde-Hofmann (LEW, s.v. *glaciēs*), misciting Lewy's page number ("anders, aber abzulehnen, Lewy KZ. 40, 119[2]"), and without bothering even to explain the nature of Lewy's proposal.

for an isolated Lat. *-iēs* form.[9] Nor is it clear how a pre-Latin **gel-ak-* would have undergone "contraction" (to use Pictet's terminology) to **glak-*. It may be possible, however, to use this account as a starting point for a more attractive explanation.

3 A new(ish) approach

Since Lat. *glaciēs* is evidently not based on the reconstructible PIE word for 'ice',[10] we must assume that the word is based on an original periphrasis or metaphorical expression, like many other words for 'ice' in IE languages (these being "mostly of obscure root connection", though in some cases "the underlying notion was that of hard or solid surface"; Buck 1949: 69). Encouraged by the similarity of *-aciēs* (in Lat. *glaciēs*) to Lat. *aciēs* itself, as well as the possibility (shared with the compound conception in 2.2 above) that a word for 'ice' might plausibly originate in a two-part expression combining 'cold' or 'frost' with 'sharpness' or 'keenness', I propose that *glaciēs* results not from a compound, but rather from a *univerbation*.[11]

It remains to specify the first member of the univerbated phrase and the phonological developments leading to the actual form *glaciēs*. To elucidate the first member, we must return to the basic words for 'cold, frost, frosty weather' cited at the outset. Attested earliest is the masc. *u*-stem *gelus* (nom. *gelus* Cato+, acc. *gelum* Cato+, abl. *gelū* Liv. Andr.+). But a neut. *o*-stem *gelum* is also found relatively early (Varro, Lucr.). Moreover, as Nussbaum has shown (1999: 401), the isolated *-idus* adjective *gelidus* is based on an adjectival abstract **gól-i-*/**gél-i-*,[12] which in turn was derived from an adjective **gel-o-* 'cold', substantivized as the *o*-stem neut. *gelum*. Thus, despite the superficial facts of attestation, *o*-stem **gel-o-* has a good claim to antiquity. Now for a univerbated two-part expression "X + *aciēs*" that will end up as *glaciēs*, "X" must end in a vowel, whence *glaciēs* as an elision product. The basis of "X", then, must be the *o*-stem **gel-o-*; and for **gel-o-*, the best-attested case form is gen. sg. *gelī* (Lucr. 5.205, 6.156, 6.530).[13] That is precisely what would be most appropriate for the original two-part expression in question, i.e., (in the manner of a kenning) **gelī aciēs* 'the sharpness/keenness of cold' = 'ice',[14] to operate with the original abstract meaning of a form like *aciēs*;[15] but one could also operate (a little more in the man-

9 This is distinct from a case like *pauperiēs* 'poverty' (XII Tab., Pl.+), evidently formed within Latin from *pauper* 'poor' (Pl.+), which happens to be a historical compound. Otherwise, compound *-iēs* forms exclusively involve preverb first members (see the listing at Bader 1962: 440).

10 Based on a PIE root **ǵeg-* (Hitt. *eka-*, OE *gicel*, etc.), Mallory and Adams 2006: 126; see also ibid. for another root (**h₁eiH-(s-)* that may have meant 'hoarfrost' (OE *īs*, Av. *aēxa-*, etc.).

11 Full disclosure: I was very likely primed to think in these terms by my recent univerbation accounts for Lat. *simītū* (Vine 2017) and Fal. **foied**/Lat. *hodiē* (Vine 2019).

12 Cf. **gól-i-* > Gmc. **kali-* in OE *ciele* 'chill' (cited above, 2.1).

13 Both **gel-o-* and **gel-u-* should have ended up with "*l* pinguis" (see Weiss 2020: 69-70, 127), whence expected †*golum* and †*golus*. The *e*-vocalism in *gelum* and *gelus* is generally taken to be analogical after the *gel-* of *gelidus* (Sihler 1995: 41, Nussbaum 1999: 387), but it seems possible that *o*-stem gen. sg. *gelī* could also have played a role.

14 On actual kennings for 'ice' in Old Icelandic poetry, see Solvin 2015: 63.

15 Piwowarczyk 2016: 115.

ner of a kenning) with the concretized sense 'sharp edge, sharp point', as in attested Latin.[16]

Univerbated phrases are quite characteristic of Latin,[17] including examples with gen. sg. in the first member (the most frequent nominal case in such formations), as in *agrīcultūra* 'agriculture', *iūrisdictiō* 'administration of justice', *plēbīscītum* 'decree of the people',[18] *vērīsimilis* 'probable', and many more.[19] The operation of elision in a univerbated ...(*V*) *V*... sequence like **gel(ī) aciēs* requires little comment: cf. *magnopere* 'greatly' (< *magnō opere*), *paenīnsula* 'peninsula' (< *paene īnsula*), *animadvertō* 'pay attention to' (< *animum advertō*), etc.

The path from **gelaciēs* to *glaciēs* is less obvious. The first point to be made is that a sequence **gél(ī) áciēs*, with adjacent primary-stressed syllables (following the elision), would be expected to undergo the reduction of one of the primary stresses, in accordance with the typologically widespread constraint against so-called "stress clash".[20] Since, moreover, *aciēs* is the head of the noun phrase, it seems reasonable to expect the accent of **gél(ī)* to undergo the reduction.[21] Under the likely assumption that the phrasal juxtaposition became fixed prior to the operation of vowel weakening, one might expect pre-Lat. **gèláciēs* to yield **gilaciēs*, with the regular weakening treatment of the unaccented vowel, as in prepositional phrase univerbations like *īlicō* 'on the spot' (< *in locō*). Irregular phonological reductions, however, are characteristic of univerbation.[22] One can imagine, then, further ("allegro"?) reductions **gl̥(l)aciēs* (via resyllabification of *-il-* to syllabic lateral)[23] and then (with loss of the syllabicity) *glaciēs*.

I have no illusions: Parts of this account—or all of it—will be hard for some readers to swallow. But that is a regular feature of research on long-standing (and possibly hopeless) etymological problems. To put a positive spin on such a rash undertaking: it gives me an opportunity to celebrate the intellectual rigor and boldness of spirit

16 On ice and its ability to cut: e.g., Virgil, E. 10.49 (*a, tibi ne teneras glacies secet aspera plantas!*), A. 12.740–1 (*mortalis mucro glacies ceu futtilis ictu / dissiluit*).

17 See in detail Bader 1962: 295–303 (on "Juxtaposés proches des composés non dérivés") and more recently Opfermann 2016.

18 On the Early Latin gen. sg. *plēbī* (to 5th-declension *plēbēs* 'the plebeian class'), see Weiss 2020: 274.

19 Additional examples in Bader 1962: 297–8. Orthographic practice, of course, varies: such items can be spelled as two words. But see the next sentence for orthographically univerbated forms with elision.

20 On "stress clash", see Gordon 2002: 505–6 (with earlier references) and 2011 (passim).

21 This would seem to run counter to the assertion by Devine and Stephens (2006: 30) that "Latin minor phonological phrases typically consist of two words and are trochaic, that is the leftmost word is prominent". Yet the situation is far more complex. Thus for two-word phrases containing a nominal argument (2006:390–1), the unmarked trochaic pattern (in a phrase like *copias Gallorum*, to use their example) can be disrupted when certain types of focus produce "an unacceptable clash between the trochaic stressing required by the language and the iambic stressing induced by the pragmatics", whence resolutions involving prosodic movement (*Gallorum copias*). In short (and there is no space for extended discussion), their system (which does operate with an appeal to the avoidance of stress clash; also 2006:131, 490) seems to allow for the "iambic" phrasing *gèl(ī) áciēs* theorized here.

22 For recent discussion in an Indo-European context, see Hackstein 2014.

23 The notation with *-l̥(l)-* is conventional for indicating a prevocalic syllabic resonant (cf. Weiss 2020: 114 fn. 54), and does not imply a geminate.

Mark Hale has always applied to tackling so many linguistic problems, all of them of far more consequence than the one under discussion here.[24]

Abbreviations

DELL	Ernout, Alfred and Antoine Meillet. 1985. *Dictionnaire étymologique de la langue latine: histore des mots*. 4e éd. rév. Paris: Klincksieck.
EDG	Beekes, Robert S. P. 2010. *Etymological Dictionary of Greek*. Leiden: Brill.
EDLIL	de Vaan, Michiel. 2008. *Etymological Dictionary of Latin and the Other Italic Languages*. Leiden: Brill.
EDPG	Kroonen, Guus. 2013. *Etymological Dictionary of Proto-Germanic*. Leiden: Brill.
IEW	Pokorny, Julius. 1959–1969. *Indogermanisches etymologisches Wörterbuch*. 2 Bde. Bern: Francke.
LEW	Walde, Alois and Johann B. Hofmann. 1938–1954. *Lateinisches etymologisches Wörterbuch*. 2 Bde. 3. Aufl. Heidelberg: Winter.
LIV^{2}	Rix, Helmut. 2001. *Lexikon der indogermanischen Verben. Die Wurzeln und ihre Primärstammbildungen. Bearbeitet von Martin Kümmel, Thomas Zehnder, Reiner Lipp und Brigitte Schirmer*. 2. Aufl. Wiesbaden: Reichert.

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